

Intro to Artificial Intelligence and Machine Learning

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- Machine Learning is part of a broader field known as **Artificial Intelligence**.
- What is Artificial Intelligence?
 - Easy part: **Artificial**.
 - Hard part: **Intelligence**.

Biological World



Artificial World



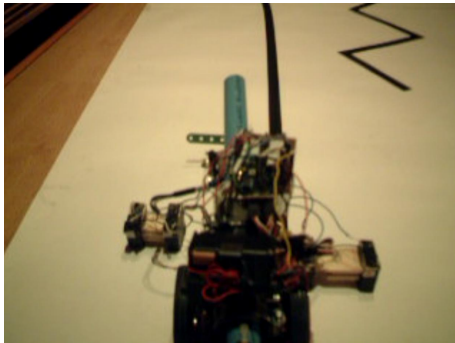
Artificial Intelligence

Study of computational models that allows machines to **perceive, reason, and act** with great **flexibility**.

Artificial Intelligence

Study of computational models that allows machines to acquire certain **level of understanding** of its world.

Understanding: Year 2001



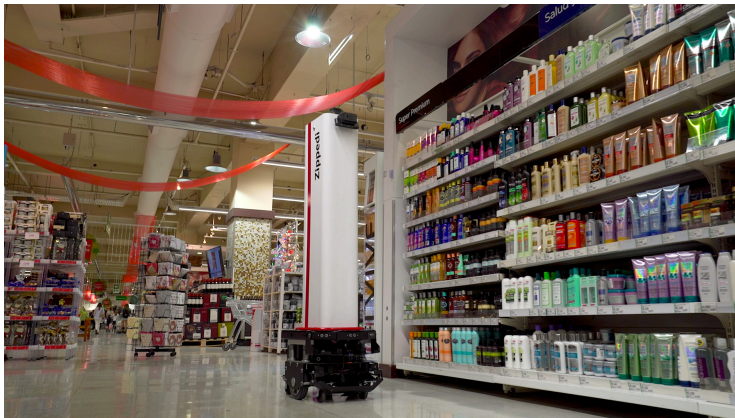
Understanding: Year 2001



Understanding: Year 2020



Understanding: Year 2020



- 60's : Era of optimism or naiveness.
 - Deductive learning.
- 70's - 80's: Era of pessimism.
- 90's: Era of reborn.
- 2000's: **Era of Machine Learning.**
 - Inductive learning.



This bird can fly



This bird can fly



This bird can fly



This bird can fly



Can this bird fly ?



This bird can fly



This bird can fly



This bird can fly



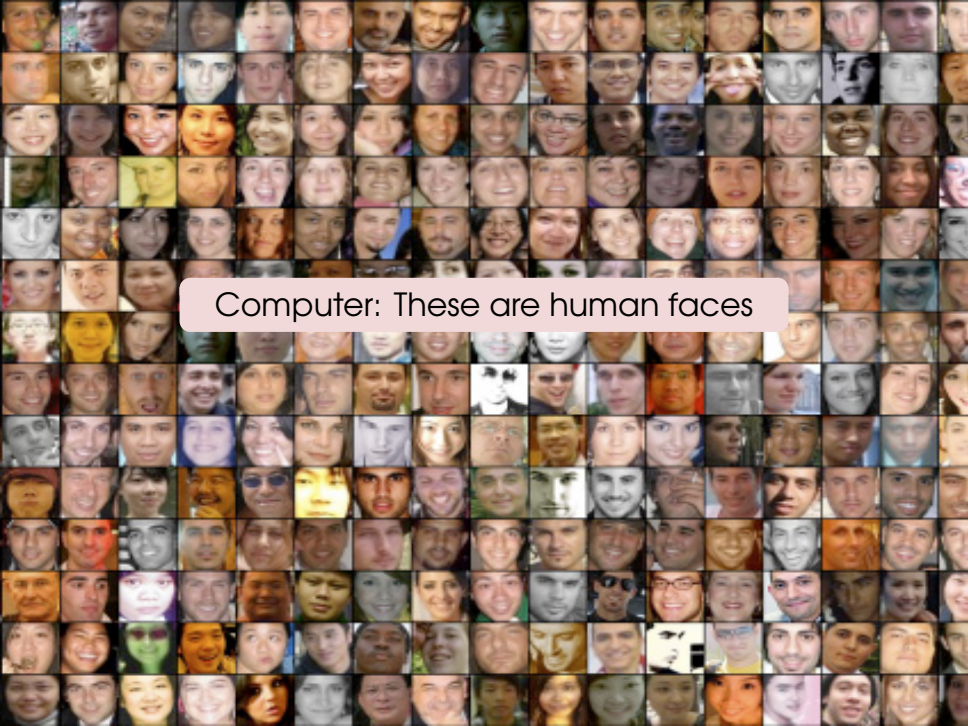
This bird can fly



Can this bird fly ?

Machine Learning

Computational programs (algorithms) that
learn from **experience**, i.e., data.



Computer: These are human faces

Computer: These are NOT human faces

Computer: Any human face?





This is a chair



This is a chair



This is a chair

Most times a
difficult problem



What are these?



AI: Modern View (Russell and Norvig)

An intelligent agent (or intelligent machine) perceives, reasons, and acts with great flexibility.

Machine learning (ML)

An intelligent agent **learns from experience**

AI: This course perspective

An intelligent agent is able to acquire certain **level of understanding** of its world.

ML: This course perspective

- Learning from experience to build **useful representations** to **understand** the world.
- Then use these representation to make prediction, take decision, etc.

- Machine learning techniques operate over multidimensional data.
- Every data example or instance is given by a set of relevant measurements or attributes.

Examples: Datasets from UCI repository
(<http://archive.ics.uci.edu/ml>):

Wine Data Set

Download: [Data Folder](#) [Data Set Description](#)

Abstract: Using chemical analysis determine the origin of wines



Data Set Characteristics:	Multivariate	Number of Instances:	178	Area:	Physical
Attribute Characteristics:	Integer, Real	Number of Attributes:	13	Date Donated	1991-07-01
Associated Tasks:	Classification	Missing Values?	No	Number of Web Hits:	573523

Bank Marketing Data Set

Download: [Data Folder](#) [Data Set Description](#)

Abstract: The data is related with direct marketing campaigns (phone calls) of a Portuguese banking institution. The class

Data Set Characteristics:	Multivariate	Number of Instances:	45211	Area:	Business
Attribute Characteristics:	Real	Number of Attributes:	17	Date Donated	2012-02-14
Associated Tasks:	Classification	Missing Values?	N/A	Number of Web Hits:	248768

Experience: Training set

The dataset used to learn a model using a machine learning technique is known as: **Training set**.

Age	Job	Marital	Education	Debt	Balance (Euros)	Housing	Loan	Contact	Day	Month	Contact duration (secs)	Campaign	Previous contacts	Subscribe deposit
30	unemployed	married	primary	no	1787	no	no	cellular	19	oct	79	1	0	no
33	services	married	secondary	no	4789	yes	yes	cellular	11	may	220	1	4	no
35	management	single	tertiary	no	1350	yes	no	cellular	16	apr	185	1	1	no
30	management	married	tertiary	no	1476	yes	yes	unknown	3	jun	199	4	0	no
59	blue-collar	married	secondary	no	0	yes	no	unknown	5	may	226	1	0	no
35	management	single	tertiary	no	747	no	no	cellular	23	feb	141	2	3	no
36	self-employed	married	tertiary	no	307	yes	no	cellular	14	may	341	1	2	no
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39	services	married	secondary	no	9374	yes	no	unknown	20	may	273	1	0	no
43	admin.	married	secondary	no	264	yes	no	cellular	17	apr	113	2	0	no
36	technician	married	tertiary	no	1109	no	no	cellular	13	aug	328	2	0	no
20	student	single	secondary	no	502	no	no	cellular	30	apr	261	1	0	yes
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31	services	married	secondary	no	132	no	no	cellular	7	jul	148	1	1	no
38	management	divorced	unknown	no	0	yes	no	cellular	18	nov	96	2	0	no
42	management	divorced	tertiary	no	16	no	no	cellular	19	nov	140	3	0	no
44	services	single	secondary	no	106	no	no	unknown	12	jun	109	2	0	no
44	entrepreneur	married	secondary	no	93	no	no	cellular	7	jul	125	2	0	no
26	housemaid	married	tertiary	no	543	no	no	cellular	30	jan	169	3	0	no
41	management	married	tertiary	no	5883	no	no	cellular	20	nov	182	2	0	no

Experience: Training set

Attributes, dimensions,
variables, or features.

Class or Label

Examples, registers, or instances.

Age	Job	Marital	Education	Debt	Balance (Euros)	Housing	Loan	Contact	Day	Month	Contact duration (secs)	Campaign	Previous contacts	Subscribe deposit
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41	management	married	tertiary	no	5883	no	no	cellular	20	nov	182	2	0	no

- Every training example can be considered as a vector that lies in a **feature space**.

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A_{14} : Pr.Cont.

A_2 : Job

A_1 : Age

Generalization: The ultimate goal

Training set used during the learning process.

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New instances **unknown** to the model.

20	student	single	secondary	no	502	no	no	cellular	30	apr	261	1	0	?
31	blue-collar	married	secondary	no	360	yes	yes	cellular	29	jan	89	1	1	
40	management	married	tertiary	no	194	no	yes	cellular	29	aug	189	2	0	
56	technician	married	secondary	no	4073	no	no	cellular	27	aug	239	5	0	

Can we measure generalization capabilities?

Training set.

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Test Set: instances with known label, but not included during training. They are used to evaluate generalization capabilities of the resulting model (why?).

20	student	single	secondary	no	502	no	no	cellular	30	apr	261	1	0	
31	blue-collar	married	secondary	no	360	yes	yes	cellular	29	jan	89	1	1	
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56	technician	married	secondary	no	4073	no	no	cellular	27	aug	239	5	0	

- One needs data to train a ML algorithm: **training set**.
- Also, in most cases, one needs to measure the performance that can be expected from a ML model: **test set**.
- Rule of thumb: from the available data, reserve 70-80% for training and 30-20% for test.

Validation set

- Machine learning techniques usually need to adjust structural parameters. Ex.: number of iterations, model structure, etc.
- To adjust these structural parameters, one needs data that is independent from the training and test sets.
- This data is known as: **validation set**.
- Rule of thumb: from the available data, reserve 60-80% for training, 10-20% for test, and 10-20% for validation.

Case example: MNIST dataset

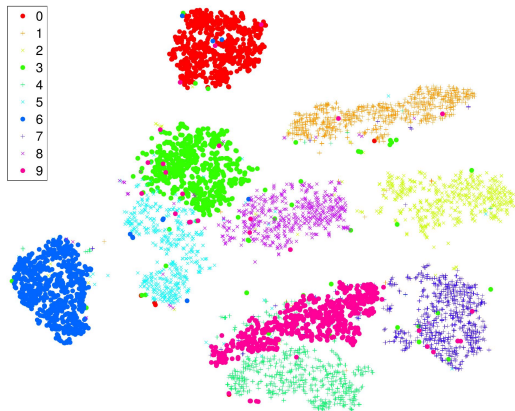


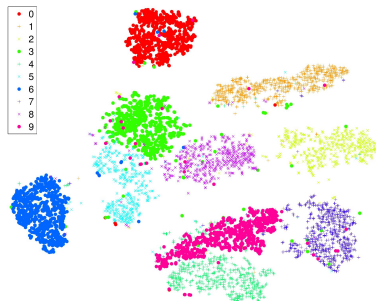
- MNIST is a dataset of images corresponding to handwritten digits.
- Each image displays a single digit from 0 to 9. Images are binary with a resolution of 28x28 pixels.
- Goal: Build a classifier that can recognize the digit in each image.
- Dataset consists of 60.000 training examples and 10.000 test cases.

What size is the feature space of the MNIST dataset?

Ej. MNIST dataset

- It is not possible to provide a direct representation of the MNIST feature space using a 2D visualization.
- However, we can use dimensionality reduction techniques such as t-SNE (<http://lvdmaaten.github.io/tsne>) to obtain a suitable approximation:





- This visualization illustrates the complexity of the target classifier.

Resulting confusion matrix after applying a K-nearest neighbors classifier over the MNIST test set.

Prediction

Real

	0	1	2	3	4	5	6	7	8	9
0	972	1	1	0	0	1	3	1	0	0
1	0	1129	3	0	1	1	1	0	0	0
2	7	6	992	5	1	0	2	16	3	0
3	0	1	2	970	1	19	0	7	7	3
4	0	7	0	0	944	0	3	5	1	22
5	1	1	0	12	2	860	5	1	6	4
6	4	2	0	0	3	5	944	0	0	0
7	0	14	6	2	4	0	0	992	0	10
8	6	1	3	14	5	13	3	4	920	5
9	2	5	1	6	10	5	1	11	1	967

These results are based on $K=1$ using Euclidean distance in the space of 784 dimensions (why 784?).

Structured data

Data that is well organized (SQL stuff).

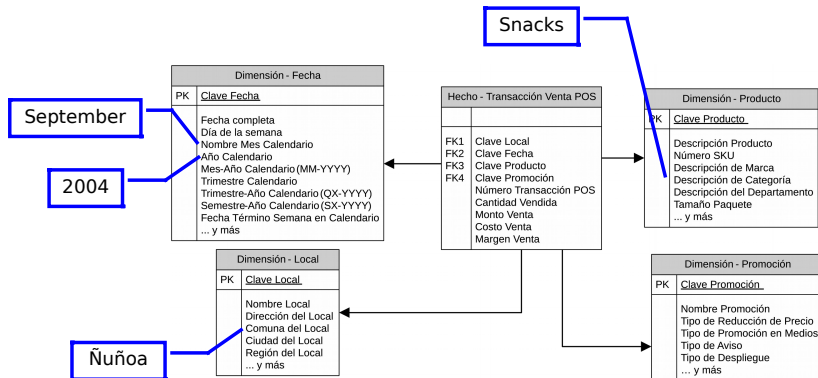
```
SELECT *  
FROM Book  
WHERE price > 100  
ORDER BY title;
```

Unstructured data

Data that is complex and not well organized.



Computer: snacks in Ñuñoa, September 2004?



Computer: What's going on?



Structured vs. Unstructured data

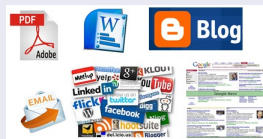
Structured data

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Unstructured data

Data that is complex and not well organized.



In simpler terms:

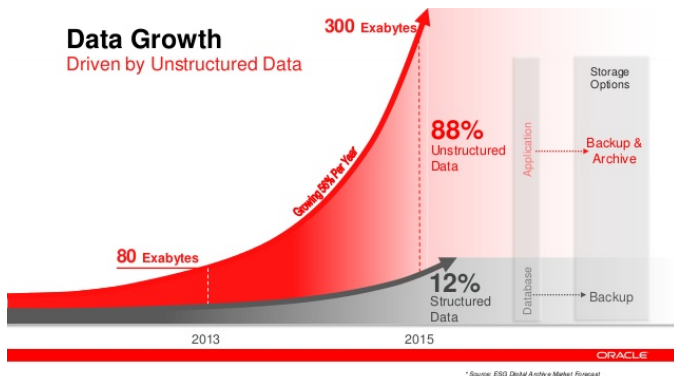
Structured data

Data that **today's computers can** understand.

Unstructured data

Data that **today's computers can't** understand.

Structured vs. Unstructured data

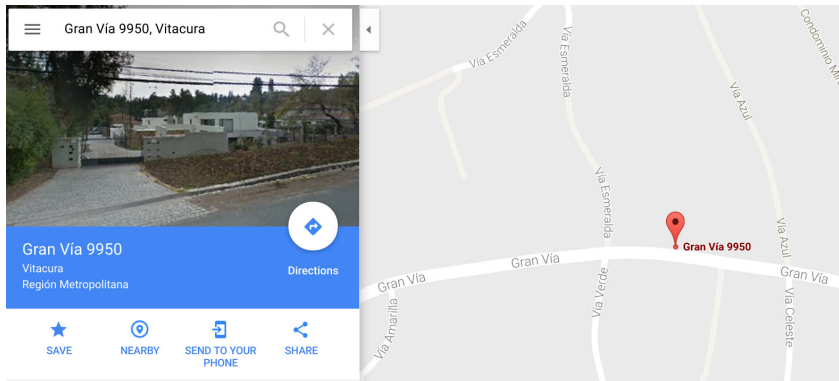


- As we discussed, Machine Learning is about learning from data.
- Currently, Big Data is it's great ally.
- By providing large sources of data, Big Data is increasing the accuracy of machine learning techniques.
- The synergistic combination between machine learning and big data is a key element behind the suces of deep learning.

Good experience, i.e. data, is key to learn.

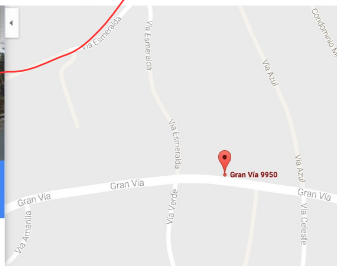
Experience is key

Example: Google Street View.



Experience is key

Example: Google Street View.



How can they do this?

Experience is key

Example: Google Street View.

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Enter the text you see in the box. Doing so helps us to prevent automated programs from abusing this service. If you cannot read the text, click the reload image  next to the text.



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This email will also contain further instructions on password resetting.*

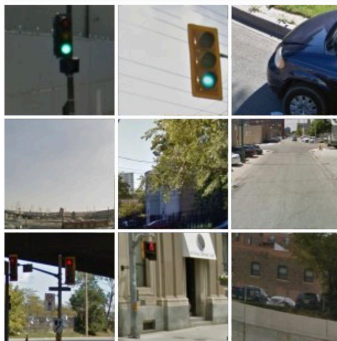
Experience is key: Training Data



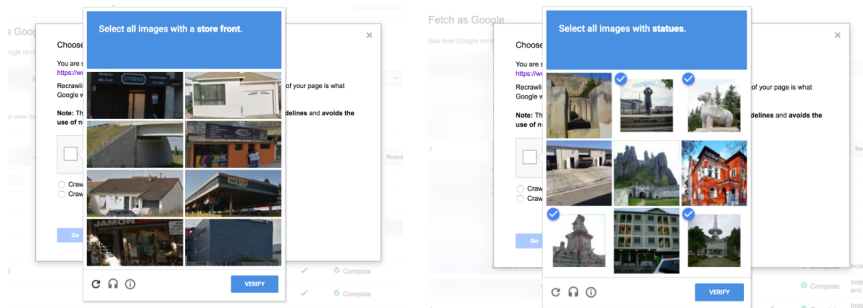
<http://research.google.com/pubs/pub42241.html>

Experience is key: **Training Data**

Select all images with
traffic lights



Experience is key: Training Data



Experience is key: Crowdsourcing and Human Computation

← → ↻ 🏠 🔒 <https://www.mturk.com/mturk/welcome> ☆ 🔍

amazonmechanical turk
Artificial Intelligence

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Mechanical Turk is a marketplace for work.
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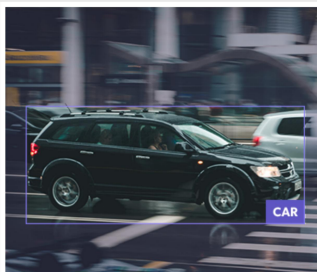
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Experience is key: Crowdsourcing and Human Computation

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FROM THE BLOG: How hCaptcha Calculates Rewards



Cost-effective data labeling at scale

Do you have large datasets you need to understand at a human level? hCaptcha provides fast, cost-effective, and high quality data labeling for AI & machine learning companies among many others.

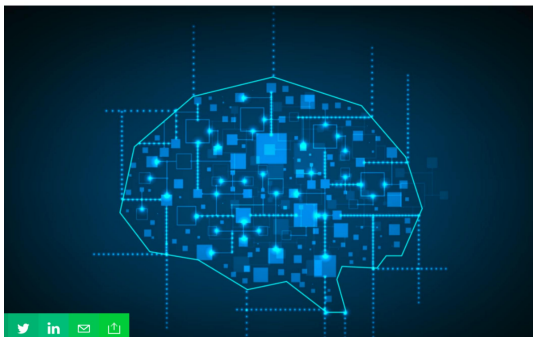
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Experience is key: Crowdsourcing and Human Computation

Appen acquires Figure Eight for up to \$300M, bringing two data annotation companies together

Anthony Ha @anthonyha / 4 hours ago

 Comment



10/03/2019

Experience is key: Training Data

Example: Xbox human pose recognition.

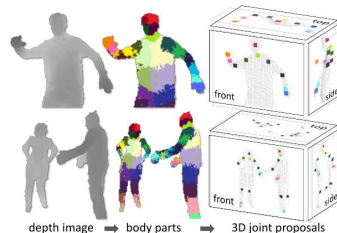
Real-Time Human Pose Recognition in Parts from Single Depth Images

Jamie Shotton Andrew Fitzgibbon Mat Cook Toby Sharp Mark Finocchio
Richard Moore Alex Kipman Andrew Blake
Microsoft Research Cambridge & Xbox Incubation

Abstract

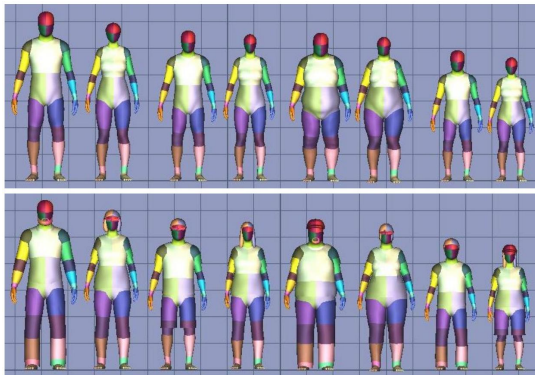
We propose a new method to quickly and accurately predict 3D positions of body joints from a single depth image, using no temporal information. We take an object recognition approach, designing an intermediate body parts representation that maps the difficult pose estimation problem into a simpler per-pixel classification problem. Our large and highly varied training dataset allows the classifier to estimate body parts invariant to pose, body shape, clothing, etc. Finally we generate confidence-scored 3D proposals of several body joints by reprojecting the classification result and finding local modes.

The system runs at 200 frames per second on consumer



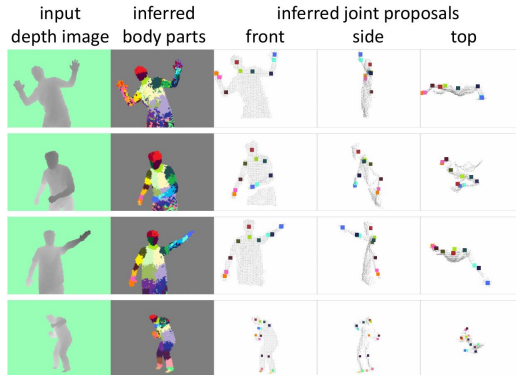
Experience is key: **Training Data**

Example: Xbox human pose recognition.



Experience is key: Training Data

Example: Xbox human pose recognition.



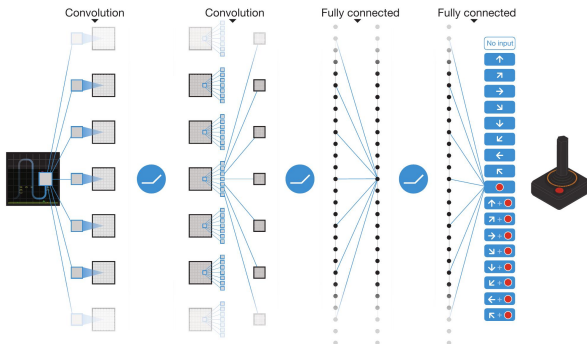
So far we have been focusing on a learning paradigm known as: **Supervised learning**. However, depending of the type of training data available and the goals of the learning process, we can find several popular learning paradigms:

- Supervised learning.
- Unsupervised learning.
- Reinforcement learning.
- Semi-supervised learning.
- Active learning.
- Structural learning.
- Supervised clustering.
- Instance based learning.
- ...

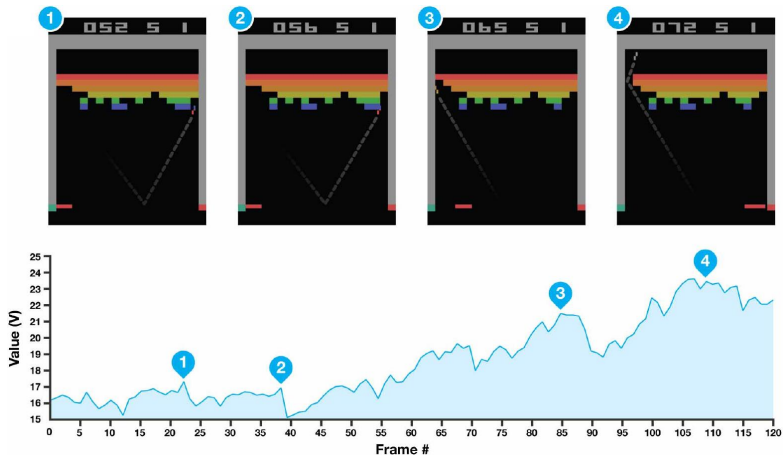
As we will discuss in this course, all these learning frameworks are just different ways to provide semantic to the learning process (grounding).

Reinforcement learning

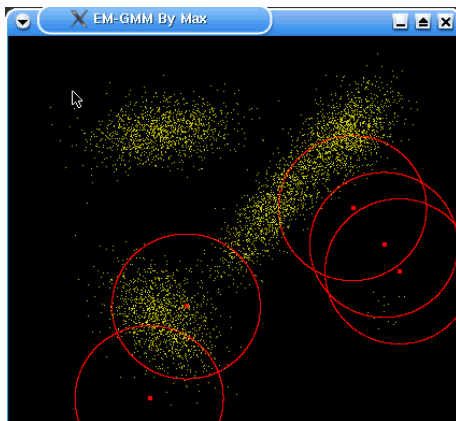
Learning to play Atari games (Mnih et al., NIPS-2014)



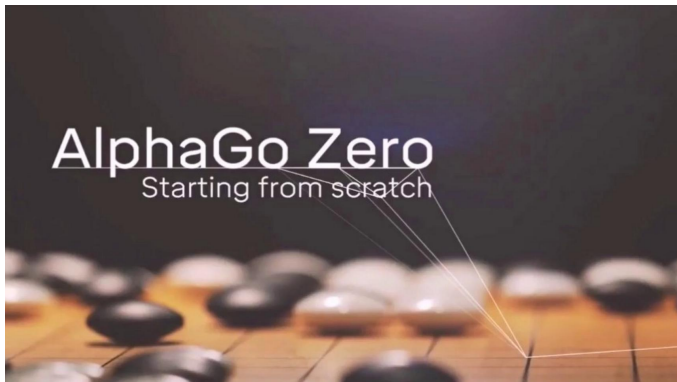
Reinforcement learning



Unsupervised Learning



Self-supervised Learning



Machine Learning

Any computer program that improves its **performance** at some **task** through **experience**.

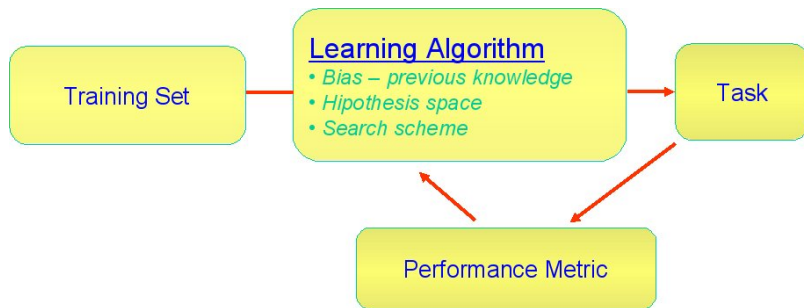
A computer program is said to learn from **experience E** with respect to some class of **task T** and **performance measure P**, if its performance for tasks in T, as measured by P, improves with experience E.

Key assumption of Inductive Machine Learning

Any hypothesis that approximates a target function well over a sufficiently large set of training examples will also approximate this target function well **over unobserved examples**.

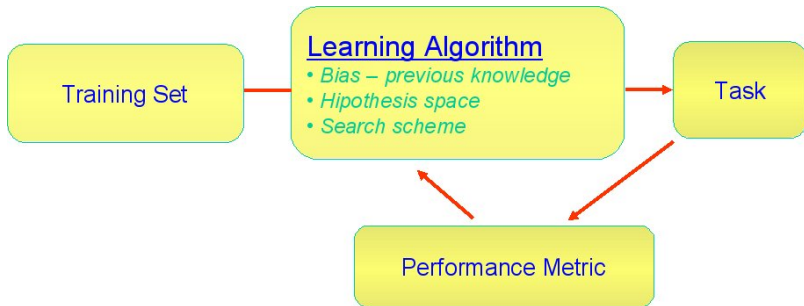
- OJO The previous assumption requires that the training data **must be representative** of the target concept or function.

Learning involves searching through a space of possible hypotheses to find a hypothesis that provides a suitable (best) fit to the available training data and prior constraints (previous knowledge).



OJO There is a new element: **previous knowledge**...Any comment?

Machine learning, 3 main ingredients:
Representation + Performance + Optimization.



Machine learning, 3 main ingredients:
Representation + Performance + Optimization.

Machine learning problem

$$f^* = \arg \min_{f \in \mathcal{H}} \mathcal{L}(f(x)) = \arg \min_{f \in \mathcal{H}} \int_{x_i \in T} \mathcal{L}(f(x_i)) d_T$$

We usually approximate this using a training set Tr :

$$f^* \approx f_{Tr}^* = \arg \min_{f \in \mathcal{H}} \frac{1}{N} \sum_{x_i \in Tr} \mathcal{L}(f(x_i))$$

$\mathcal{H}$: hypothesis space.

$\mathcal{L}$: loss function

f^* : optimal hypothesis in $\mathcal{H}$ under $\mathcal{L}$.

Generic machine learning loss

$$f^* \approx f_{Tr}^* = \arg \min_{f \in \mathcal{H}} \frac{1}{N} \sum_{x_i \in Tr}^N \mathcal{L}(f(x_i))$$

Supervised learning

$$f_{Tr}^* = \arg \min_{f \in \mathcal{H}} \frac{1}{N} \sum_{x_i, y_i \in Tr}^N \mathcal{L}(f(x_i), y_i)$$

- Models have limits (suitable representation?).
- Search in complex spaces (non-linear or convex) is usually a hard and exhausting process (right optimization tool and performance metric?).

Risks,

- If the hypothesis space is too flexible, there is a higher risk to converge to misleading hypothesis (suffer overfitting problems).
- If the search tool is too limited, there is a higher risk to being unable to find a good hypothesis (converge to local optimals).

Overfitting?

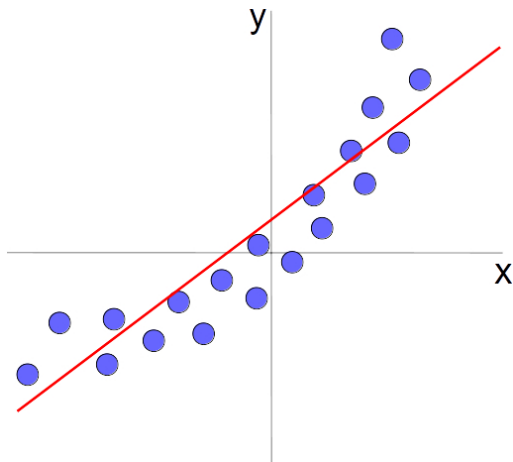
- When one applies a ML technique, one is interested in the generalization capabilities of the resulting model, why?.
- Generalization? → **Inductive learning**.
- In other words, one is interested in obtaining good predictions for new instances of the target domain.

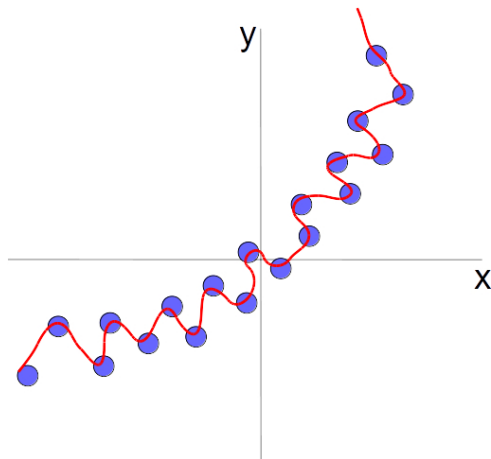
Overfitting: It is a situation when a ML model starts to memorize (overfit) the training data, decreasing its capability to correctly predict new instances.

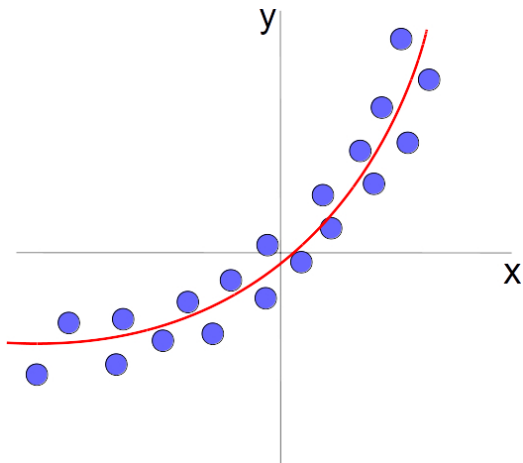
Jorge Luis Borges: “Funes el memorioso”.

“...Había aprendido sin esfuerzo el inglés, el francés, el portugués, el latín. Sospecho, sin embargo, que no era muy capaz de pensar. Pensar es olvidar diferencias, es **generalizar, abstraer**. En el abarrotado mundo de Funes no había sino detalles, casi inmediatos.”

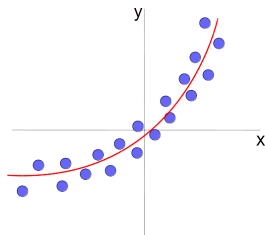
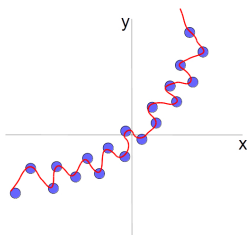
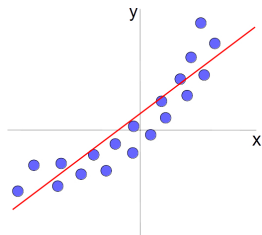
Jorge Luis Borges: “Funes el memorioso”.

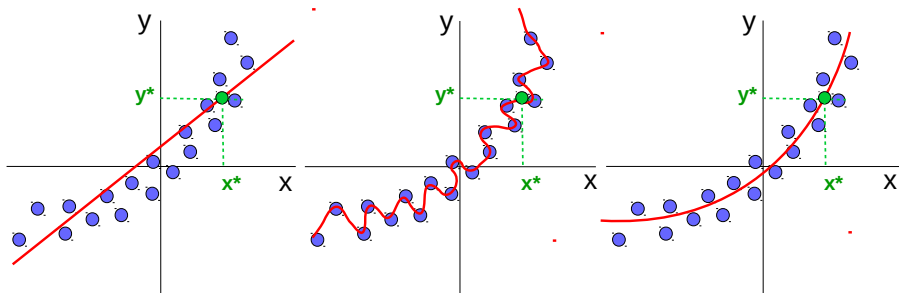


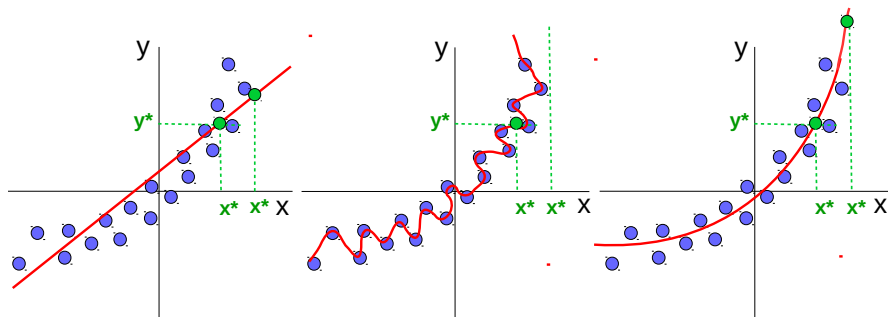




Underfitting, Overfitting, Good fit







Concepts that you should know about ML.

- Classical AI and ML views.
- Generic view of ML.
- Main elements of a ML technique: Representation + Performance + Optimization.
- Supervised and unsupervised ML.
- Overfitting.
- Training, test, validation sets.

Recommended concepts that you should review. We will discuss some of them in future lectures:

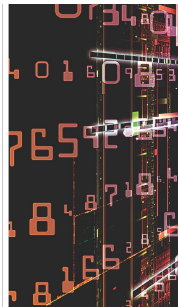
- All the ideas in this week lecture: “A Few Useful Things to Know About Machine Learning” by Pedro Domingos.

Tapping into the “folk knowledge” needed to advance machine learning applications.

BY PEDRO DOMINGOS

A Few Useful Things to Know About Machine Learning

MACHINE LEARNING SYSTEMS automatically learn programs from data. This is often a very attractive alternative to manually constructing them, and in the last decade the use of machine learning has spread rapidly throughout computer science and beyond. Machine learning is used in Web search, spam filters, recommender systems, ad placement, credit scoring, fraud detection, stock trading, drug design, and many other applications. A recent report from the McKinsey Global Institute asserts that machine learning (a.k.a. data mining or predictive analytics) will be the driver of the next big wave of innovation.¹⁵ Several fine textbooks are available to interested practitioners and researchers (for example, Mitchell¹⁶ and Witten et al.²⁴). However, much of the “folk knowledge” that



is needed to successfully develop machine learning applications is not readily available in them. As a result, many machine learning projects take much longer than necessary or wind up producing less-than-ideal results. Yet much of this folk knowledge is fairly easy to communicate. This is the purpose of this article.

» key insights

- Machine learning algorithms can figure out how to perform important tasks by generalizing from examples. This is often feasible and cost-effective where manual programming is not. As more data becomes available, more ambitious problems can be tackled.
- Machine learning is widely used in computer science and other fields. However, developing successful machine learning applications requires a substantial amount of “black art” that is difficult to find in textbooks.
- This article summarizes 12 key lessons that machine learning researchers and practitioners have learned. These include pitfalls to avoid, important issues to focus on, and answers to common questions.